Yongye(Felix) Hu

510-277-7666 | hyy.felix@gmail.com | felix-hyy | felix-hu.space

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Hiring Manager
Energize Group

Dear Hiring Manager,

I am writing to apply for the Senior Robotics and AI Research (Physical AI Architect) position. As a robotics engineer with an M.Eng from UC Berkeley, I bring hands-on depth in optimal control, ROS2-based autonomy, and real-time embedded systems—a foundation I aim to extend into learned policy architectures for warehouse automation.

At Berkeley, I developed a ROS2-based system for a hybrid flying-driving robot, integrating an MPC controller with an OptiTrack motion capture loop that delivered sub-millimeter position accuracy and under 10 ms feedback latency. Through iterative experimental validation and controller refinement, the platform achieved sub-centimeter trajectory tracking and completed full obstacle avoidance runs reliably. This end-to-end controls pipeline—from simulation to hardware validation—maps directly to the role's need for combining optimal control with physical system constraints.

My embedded and hardware/software co-design experience complements that controls work. On the ROBOCON competition platform, I wrote STM32 firmware in C for CAN-based motor control and developed a three-loop PID chassis architecture that converged within 1.2 seconds, while leading three interdisciplinary teams through a six-month build cycle. Separately, I designed a plug-and-play sensor-control module with real-time diagnostics for a vacuum dryer prototype, accelerating hardware/software debugging and reinforcing my instinct for system observability and reliability—qualities the role demands across sensors, compute, and actuation.

I want to be direct: I have not worked with diffusion-based policy learning or deployed learned policies in production. My strength is the classical controls and systems integration layer that learned manipulation and picking policies must ultimately execute through in warehouse environments. I would welcome the opportunity to discuss how that foundation can serve your autonomy stack and where I can grow into the AI side of the architecture.

Sincerely,
Felix Hu